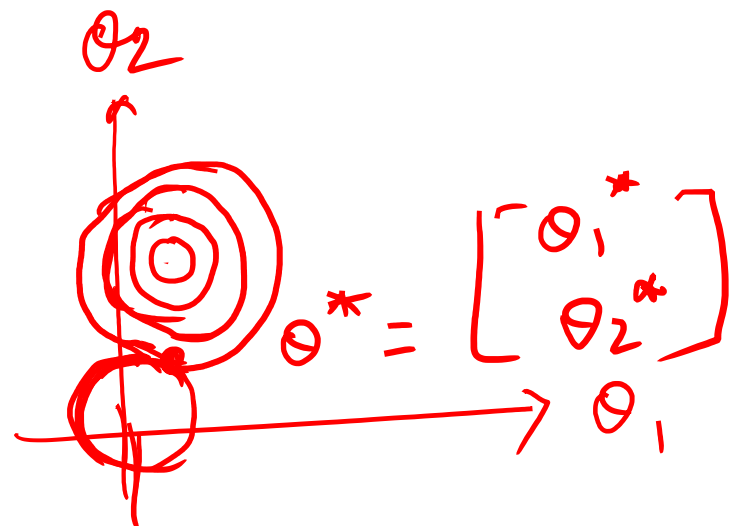
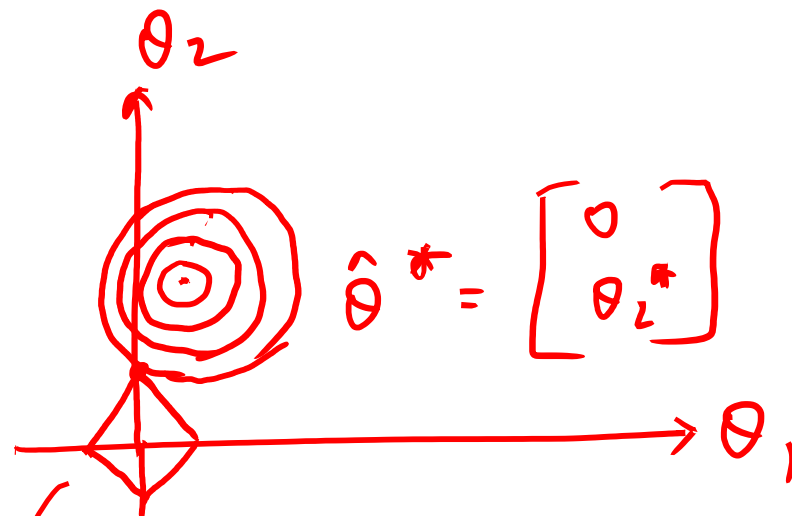


LASSO & Cross Validation



Ridge Regression

$$\|\theta\|_2^2$$



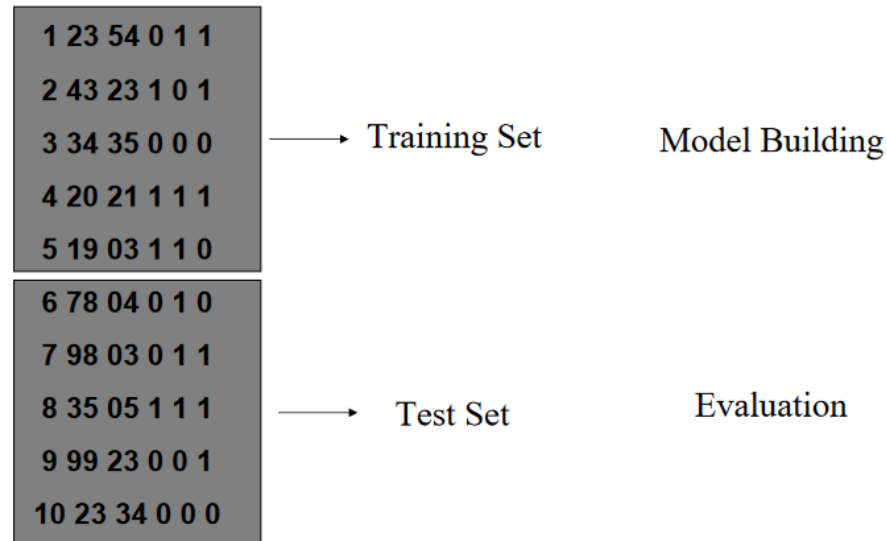
LASSO

$$\|\theta\|_1$$

Sparsity

Training and Testing Sets

- Training set is used to build the model
- Test set left aside for evaluation purposes
- Ideal: different data set to test if model generalizes to other settings
- If data are abundant, then there is no need to “recycle” cases



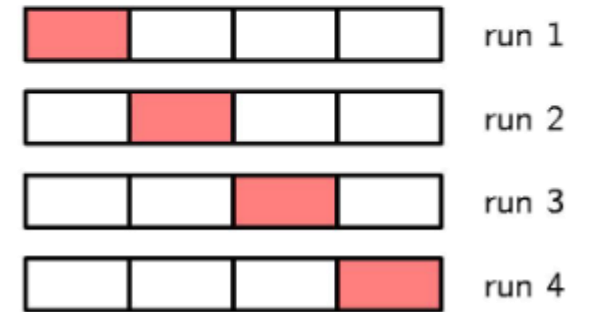
Cross-Validation

- The simplest and most widely used method for estimating prediction error is cross-validation.
- When data set is too small, dividing into a fixed training set and fixed testing set is problematic
 - If it results in a small test set
 - Small test set implies statistical uncertainty around the estimated average test error
 - Cannot claim algorithm A works better than algorithm B for a given task
- k - fold cross-validation
 - Partition the data into k non-overlapping subsets
 - On trial i , i th subset of data is used as the test set
 - Rest of the data is used as the training set

k - fold Cross-Validation

- Supply of data is limited
- All available data is partitioned into k groups (folds)
- $k - 1$ groups are used to train and evaluated on remaining group
- Repeat for all k choices of held-out group
- Performance scores from k runs are averaged

$k=4$



k - fold Cross-Validation

- $\kappa: \{1, 2, \dots, n\} \mapsto \{1, 2, \dots, k\}$ -- An indexing function that indicates the partition to which observation i is allocated by the randomization.
- The fitted function is $\hat{f}^{-k}(x)$ be the fitted function, computed with the k th part of the data removed.
- The cross-validation estimate of the prediction error is

$$CV = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}^{-\kappa(i)}(x_i))$$

- Typical choices of K are 5 or 10.
- $k = n$ is known as *leave-one-out* cross-validation. (LOOCV)

$$\hat{\theta}_1, \hat{\theta}_2, \dots$$

$$\boxed{E[\hat{\theta}] = \theta} \leftarrow \text{original parameter.}$$

↓
Estimator $\rightarrow$ unbiased.

$$E[\hat{\theta}] - \theta \neq 0 \rightarrow \text{Biased.}$$

$$E[\hat{\theta}] - \theta = \text{Bias.}$$

LOOCV :-

$$x^{(i)} = A + w^{(i)} \quad i=1, \dots, n$$

$A = \underline{\text{unknown DC level}}$

$$w^{(i)} \sim \mathcal{N}(0, \sigma^2)$$

MLE :-

$$\hat{A} = \frac{1}{n} \sum_{i=1}^n x^{(i)}$$

$$E[\hat{A}] = \frac{1}{n} \sum_{i=1}^n E[x^{(i)}]$$

Unbiased

$$= \frac{1}{n} \sum_{i=1}^n A = \frac{nA}{n} = A$$

$$x^{(i)} = A + w^{(i)}$$

$$E[x^{(i)}] = A$$

$$\text{var}(\hat{A}) = \frac{1}{n^2} \sum_{i=1}^n \text{var}(x^{(i)}) = \frac{\sigma^2}{n}$$

Leave One Out Estimator :-

$$\hat{A}_{(-k^{(i)})} = \frac{1}{n-1} \sum_{i \neq n} x^{(i)}$$

$$E[\hat{A}_{(-k^{(i)})}] = \frac{1}{n-1} \sum_{i \neq n} E[x^{(i)}] = A$$

$$\text{Var}(\hat{A}_{-k(i)}) = \frac{1}{(n-1)^2} \cdot (n-1) \sigma^2 = \frac{\sigma^2}{n-1}$$

$$\text{Var}(\hat{A}_{-k(i)}) \geq \text{Var}(\hat{A})$$

Linear Error:- $e^{(i)} = x^{(i)} - \hat{A}_{-k(i)}$

$$x^{(i)} = A + w^{(i)}$$

$$e^{(i)} = A + w^{(i)} - \frac{1}{n-1} \sum_{n \neq i} [A + w^{(i)}]$$

$$e^{(i)} = w^{(i)} - \frac{1}{n-1} \sum_{n \neq i} w^{(i)}$$

$$E[e^{(i)}] = 0$$

$$\text{Var}(e^{(i)}) = \text{Var}(w^{(i)}) + \text{Var}\left(\frac{1}{n-1} \sum_{n \neq i} w^{(i)}\right)$$

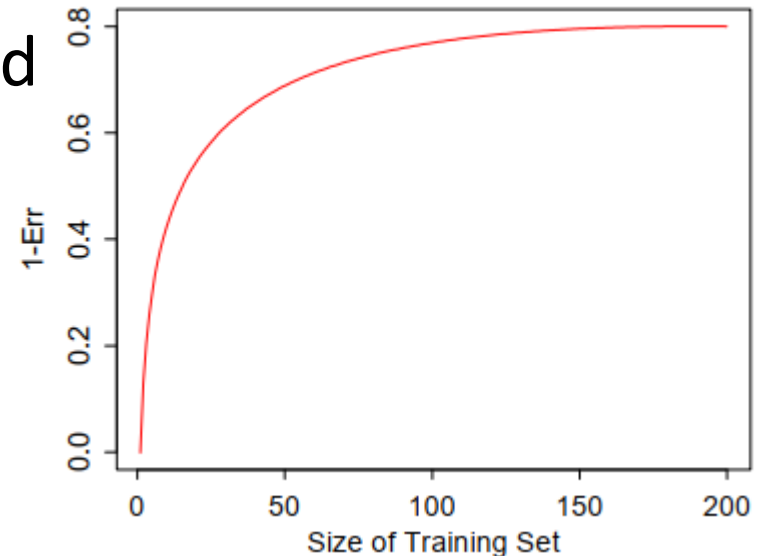
$$= \sigma^2 + \frac{\sigma^2}{n-1} = \frac{n}{n-1} \sigma^2$$

For small n , fluctuations will be noticeable.

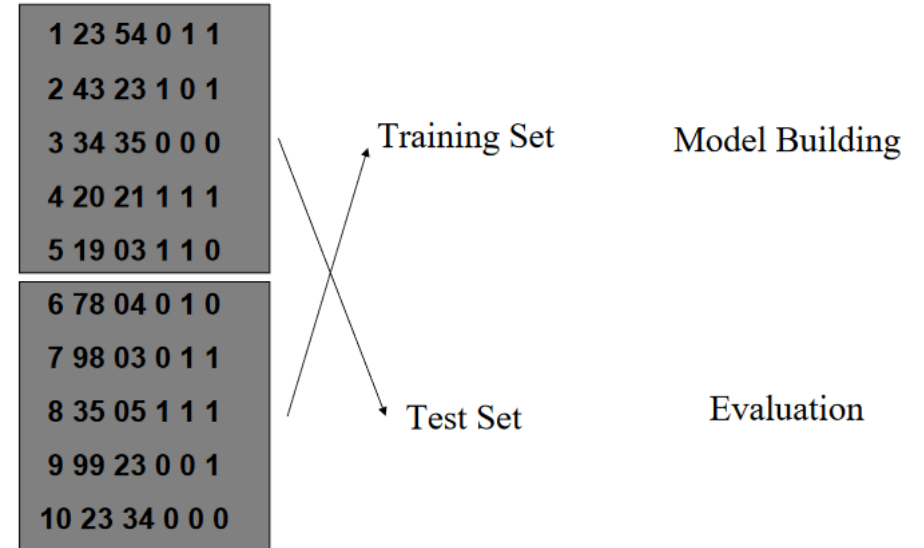
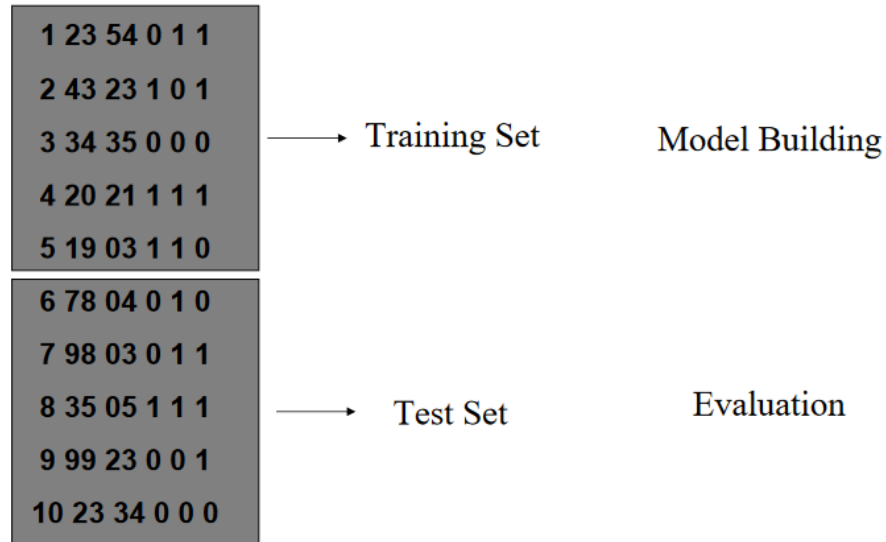
" large n , " not be "

What value should we choose for k ?

- $k = n$, CV is approximately unbiased for the true prediction error, but can have high variance because all the training sets are so similar to one another
- $k = 5$, CV shall have lower variance, but bias could be problem, depending on how the performance of the learning method varies with the size of the training set.
- For 200 observations, 5-fold cross-validation would not suffer from much bias
- For 50 observations, 5-fold cross-validation would underestimate 1-Err.



$k=2$ Example



$k=3$ Example

